Course Title: Ethical Hacking

Objective

The objective of this experiment is to perform Open Source Intelligence (OSINT) investigations using Maltego, a visual link-analysis platform. Students will learn to represent information as entities, use transforms to discover related data, and trace relationships to build a complete picture of a target's digital footprint, further enriched using public datasets and Shodan.

Methodology

1. Introduce the concept of Open Source Intelligence (OSINT) and how it applies to visual link-analysis tools.
2. Install and configure Maltego Community Edition (CE) for entity-based investigation.
3. Use Maltego transforms to discover DNS names, email addresses, and IP addresses related to a target domain.
4. Trace relationships between discovered entities to identify significant connections (e.g. shared infrastructure).
5. Enrich findings using public datasets and Shodan to cross-verify and expand the investigation.
6. Analyze the collected information to compile a documented OSINT assessment.

Expected Outcome

By completing this experiment, you will be able to:

1. Explain the core principles of OSINT and its role in ethical hacking.
2. Install and navigate the Maltego Community Edition interface.
3. Build an entity graph using domain, DNS, email, and IP transforms.
4. Trace relationships between entities to identify significant findings.
5. Use public datasets and Shodan to enrich and verify OSINT findings.
6. Compile a structured OSINT investigation report from a visual entity graph.

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1. Introduction to OSINT

Open Source Intelligence (OSINT) is the practice of collecting and analyzing publicly available information — websites, public records, and metadata — to build actionable intelligence about a target, without bypassing any access controls or interacting with private systems.

1.1. Uses of OSINT

- Attackers use OSINT to craft phishing emails and identify weak points before ever touching a target's systems.
- Defenders use the same techniques to understand and reduce their own organization's exposure.
- Every tool used in this experiment (Maltego, public datasets, Shodan) operates entirely on data that is already public.

2. Introduction to Maltego

Maltego is a visual link-analysis platform used for OSINT investigations. Instead of reading results as a flat list, Maltego represents information as connected "entities" on a graph, making relationships between findings much easier to identify.

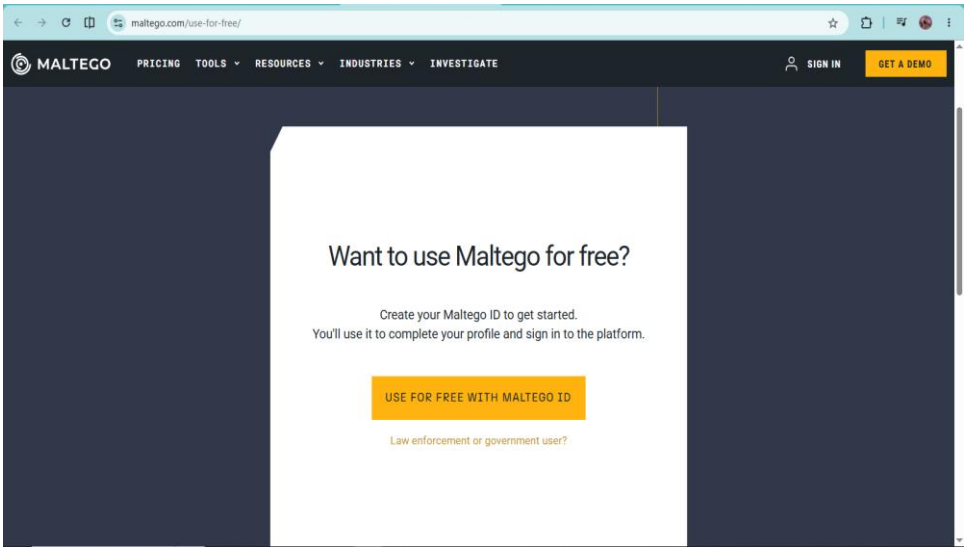
2.1. Core Concepts

1. **Entity** — a single piece of data, such as a Domain, Email Address, Person, or IP Address.
2. **Transform** — a query that takes one entity and discovers new, related entities.
3. **Graph** — the growing visual map of entities and the links between them.

3. Setting up Maltego

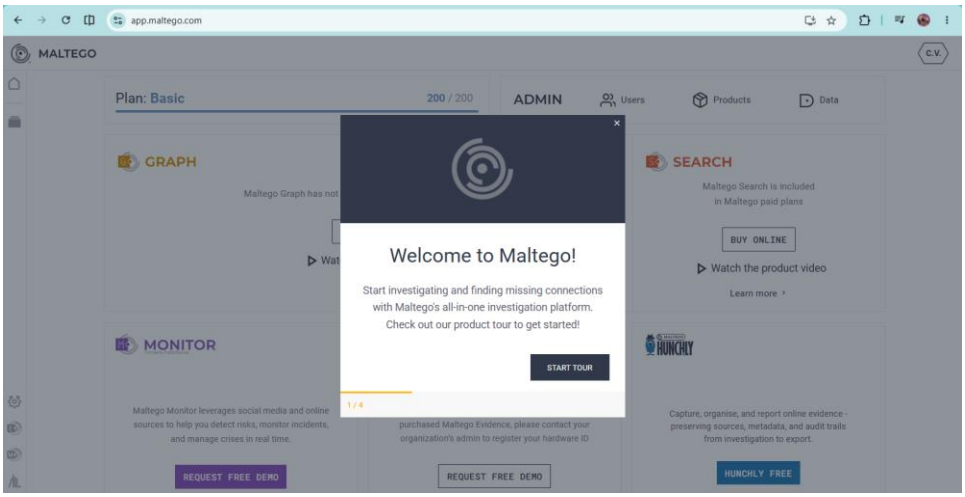
Step 1: Register a Maltego Community Edition Account

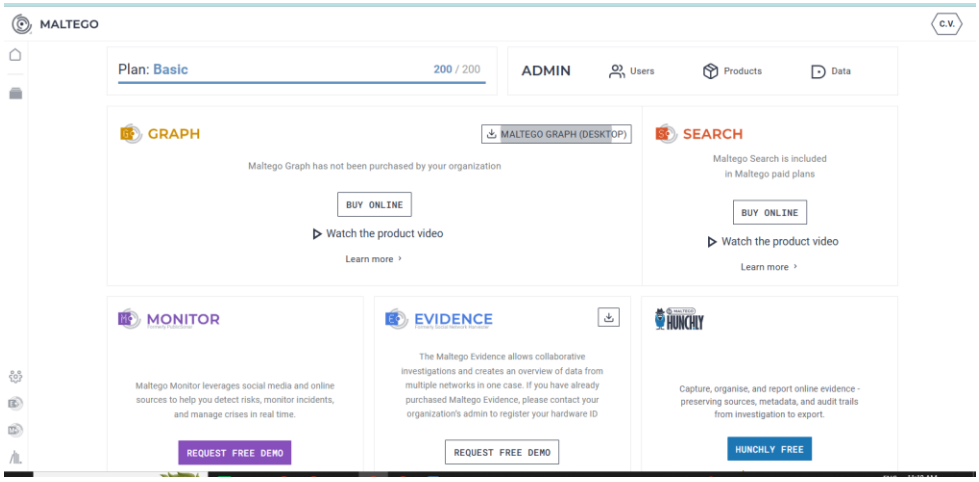
Visit maltego.com/ce-registration and create a free Community Edition account using a valid email address.



Step 2: Verify Your Email

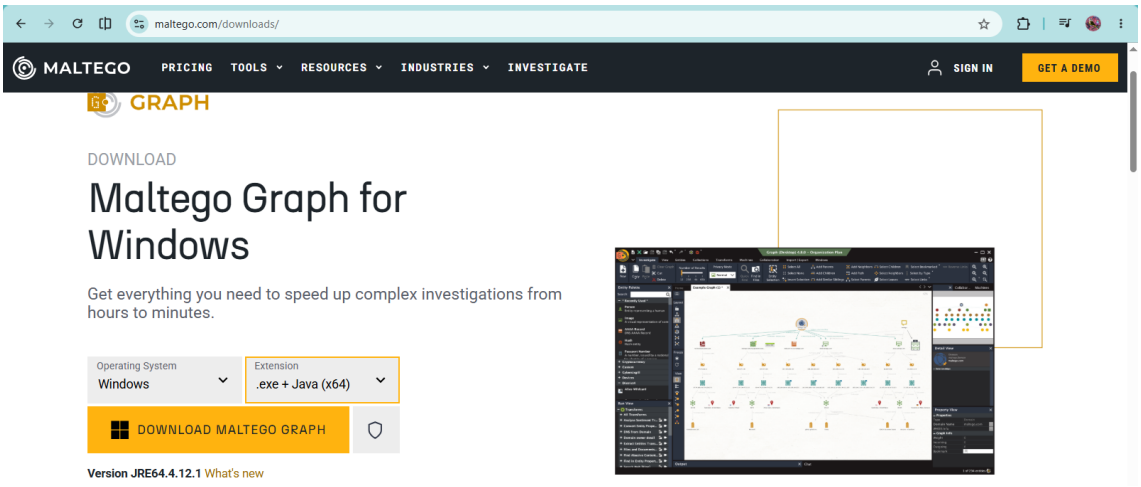
Open the confirmation email sent by Maltego and click the verification link before attempting to log in. After login click Maltego Graph(desktop) for installation.

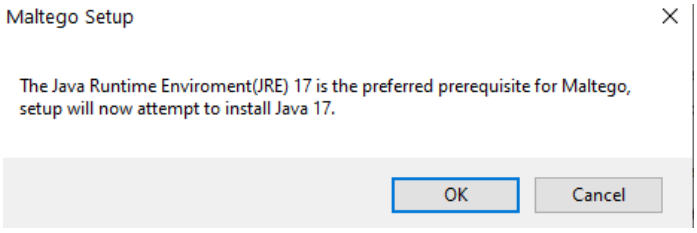


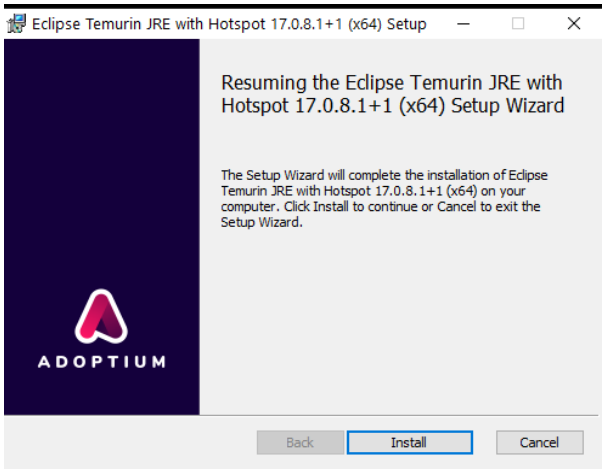


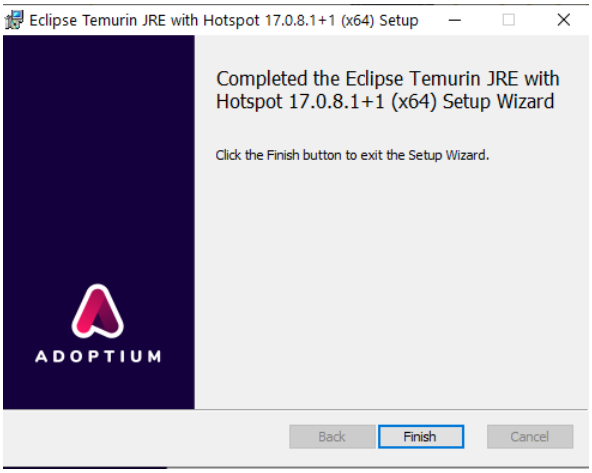
Step 3: Install and Launch Maltego

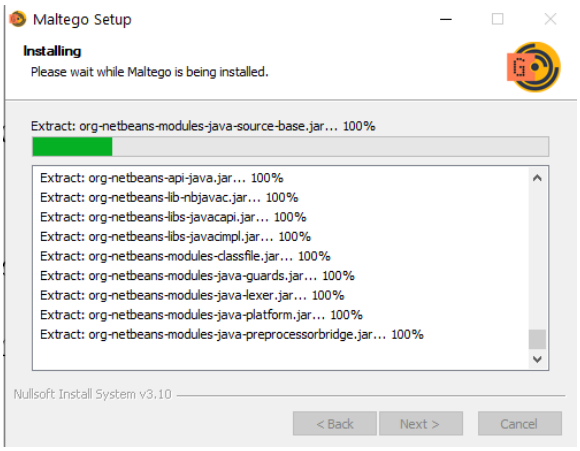
Download Maltego from the official website (or use the version pre-installed on Kali Linux) and launch the application.

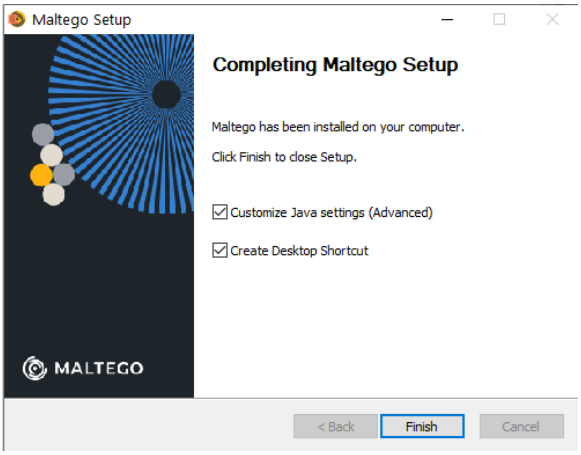

maltego.com/downloads/
MALTEGO
PRICING
TOOLS
RESOURCES
INDUSTRIES
INVESTIGATE
SIGN IN
GET A DEMO
GRAPH
DOWNLOAD
Maltego Graph for Windows
Get everything you need to speed up complex investigations from hours to minutes.
Operating System
Windows
Extension
.exe + Java (x64)
DOWNLOAD MALTEGO GRAPH
Version JRE64.4.12.1 What's new


Maltego Setup
The Java Runtime Enviroment(JRE) 17 is the preferred prerequisite for Maltego, setup will now attempt to install Java 17.
OK
Cancel


Eclipse Temurin JRE with Hotspot 17.0.8.1+1 (x64) Setup
Resuming the Eclipse Temurin JRE with Hotspot 17.0.8.1+1 (x64) Setup Wizard
The Setup Wizard will complete the installation of Eclipse Temurin JRE with Hotspot 17.0.8.1+1 (x64) on your computer. Click Install to continue or Cancel to exit the Setup Wizard.
ADOPTIUM
Back
Install
Cancel

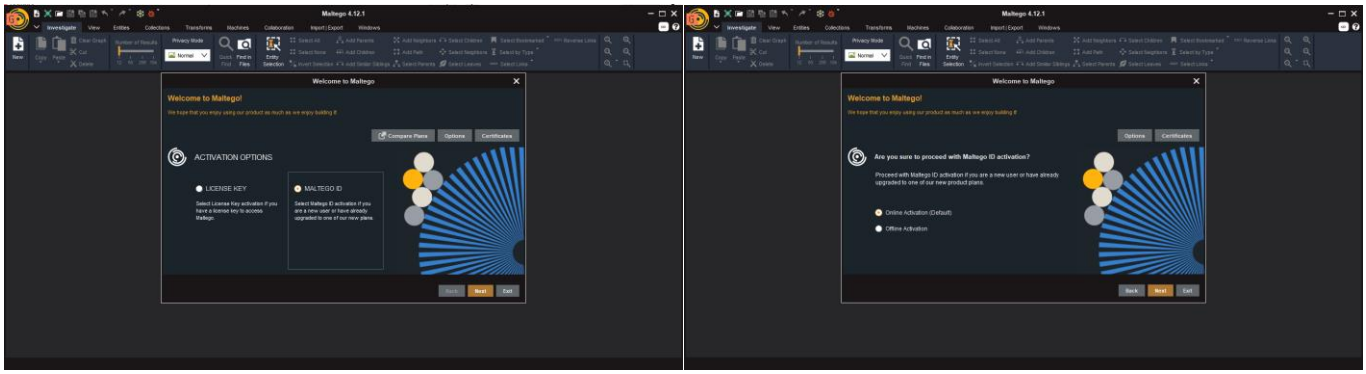

Eclipse Temurin JRE with Hotspot 17.0.8.1+1 (x64) Setup
Completed the Eclipse Temurin JRE with Hotspot 17.0.8.1+1 (x64) Setup Wizard
Click the Finish button to exit the Setup Wizard.
ADOPTIUM
Back
Finish
Cancel

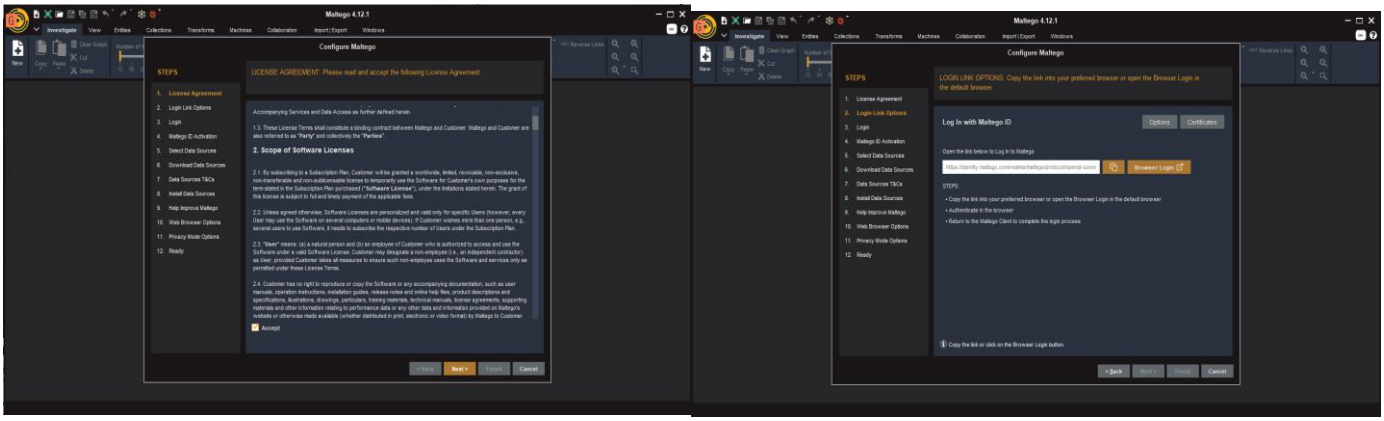

Maltego Setup
Installing
Please wait while Maltego is being installed.
Extract: org-netbeans-modules-java-source-base.jar... 100%
Extract: org-netbeans-api-java.jar... 100%
Extract: org-netbeans-lib-nbjavac.jar... 100%
Extract: org-netbeans-libs-javacapi.jar... 100%
Extract: org-netbeans-libs-javacimpl.jar... 100%
Extract: org-netbeans-modules-classfile.jar... 100%
Extract: org-netbeans-modules-java-guards.jar... 100%
Extract: org-netbeans-modules-java-lexer.jar... 100%
Extract: org-netbeans-modules-java-platform.jar... 100%
Extract: org-netbeans-modules-java-preprocessorbridge.jar... 100%
Nullsoft Install System v3.10
< Back
Next >
Cancel

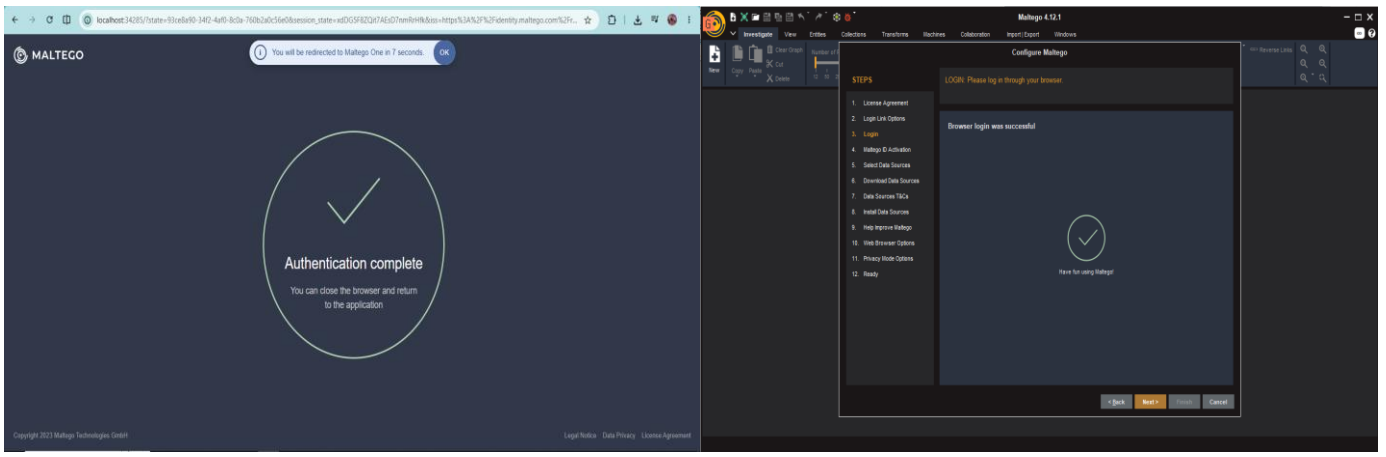

Maltego Setup
Completing Maltego Setup
Maltego has been installed on your computer.
Click Finish to close Setup.
Customize Java settings (Advanced)
Create Desktop Shortcut
MALTEGO
< Back
Finish
Cancel

Step 4: Log In

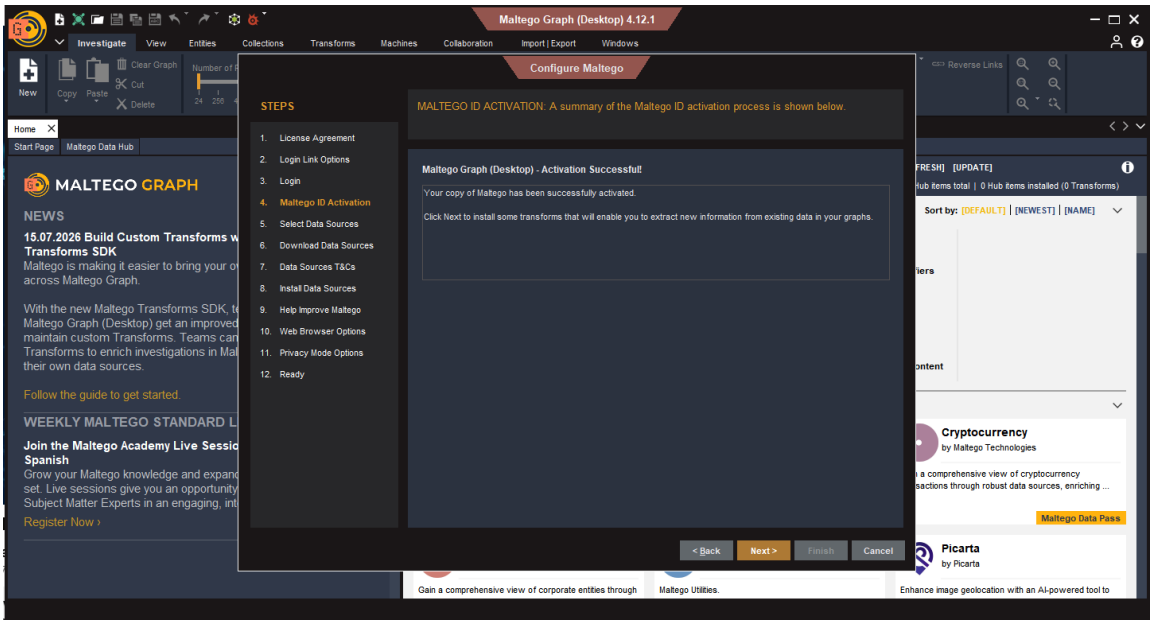
Open Maltego, choose “Log in with Maltego account”, and enter your verified credentials.



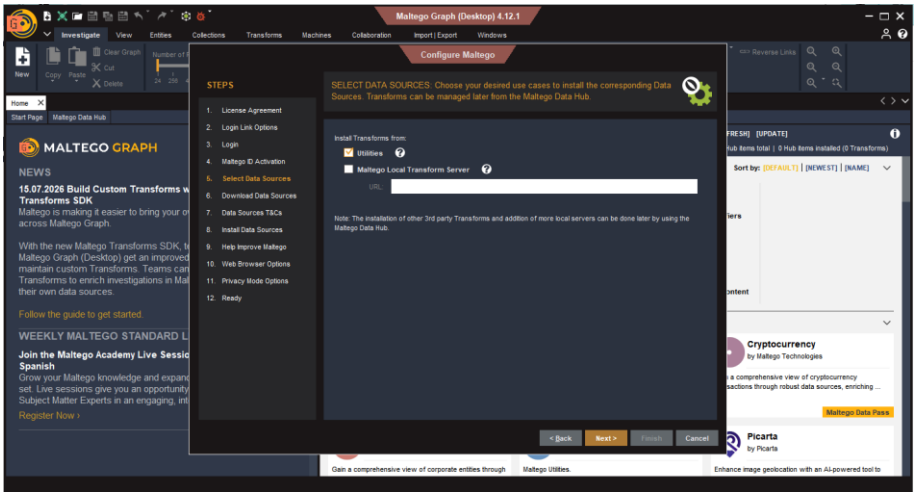




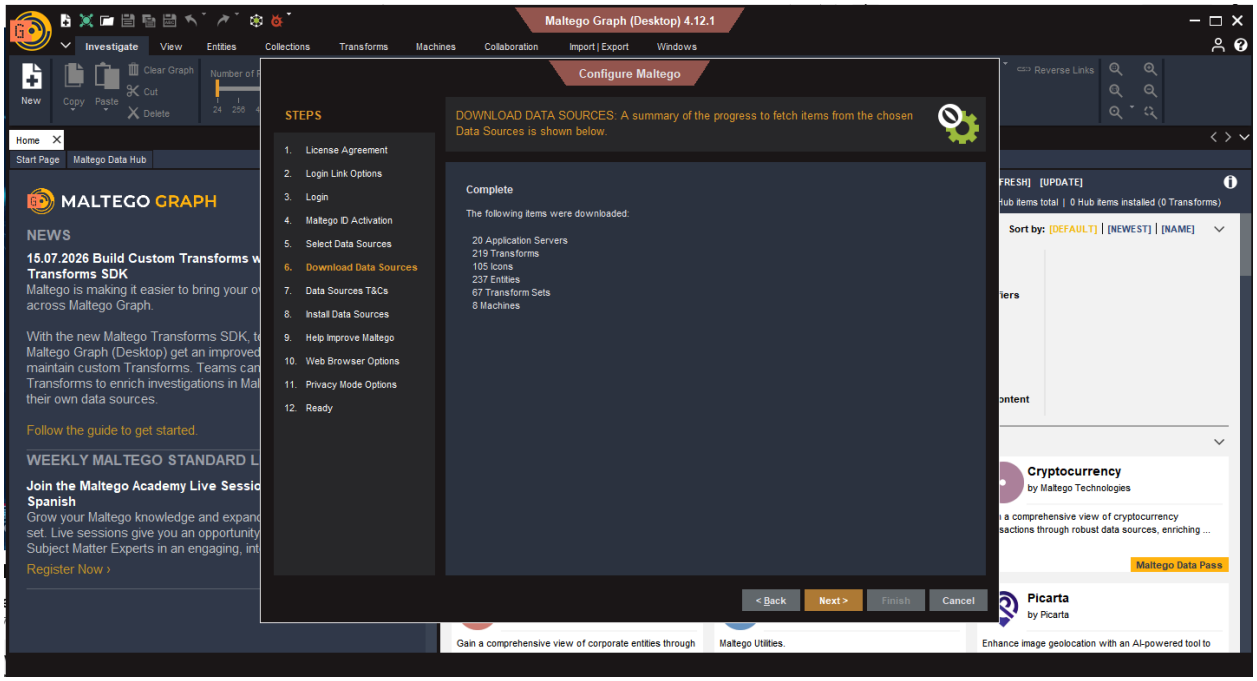
Click Next to all the process after successful Sign in



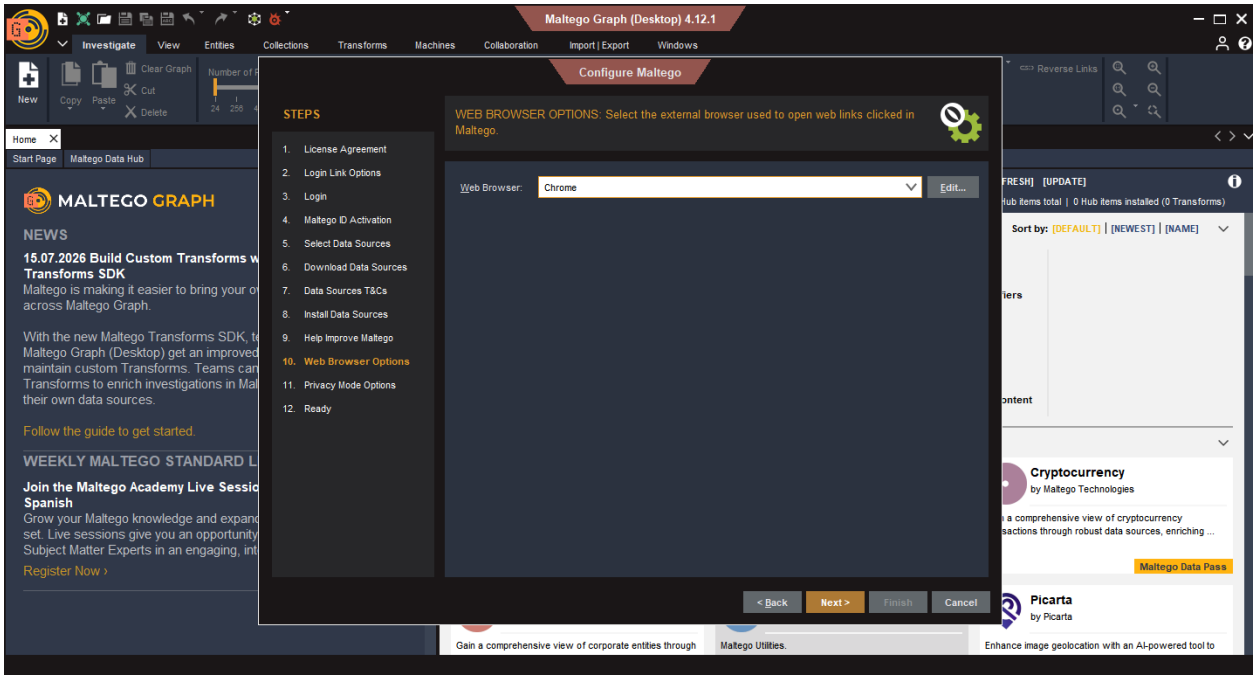
Select data source with default utilities and click next.



Download the data sources and Click Next.



Choose Your Default Browser and select Next.



Click open an example graph or a blank graph and click finish.

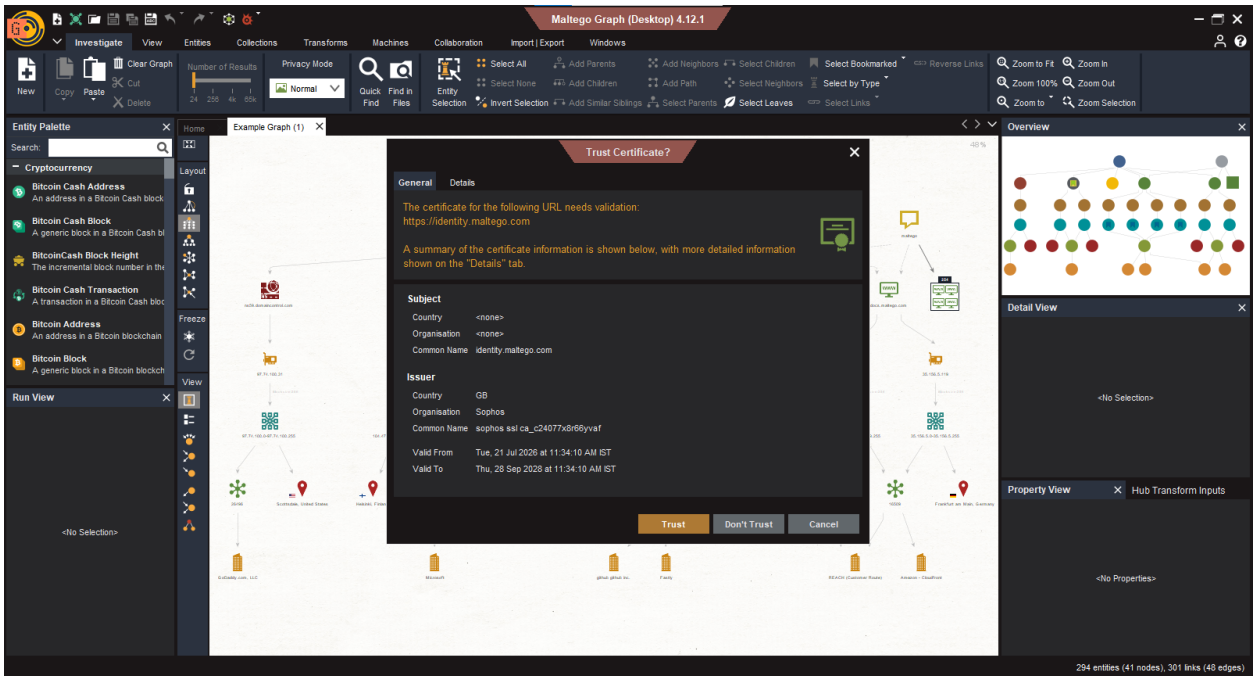
4. Maltego Interface Overview

1.Entity Palette (left panel) — drag entities onto the canvas to begin an investigation.

2.Graph Canvas (centre panel) — the main workspace where entities and their links are displayed.

3.Transform Hub / Detail View (right panel) — run transforms and inspect entity properties.

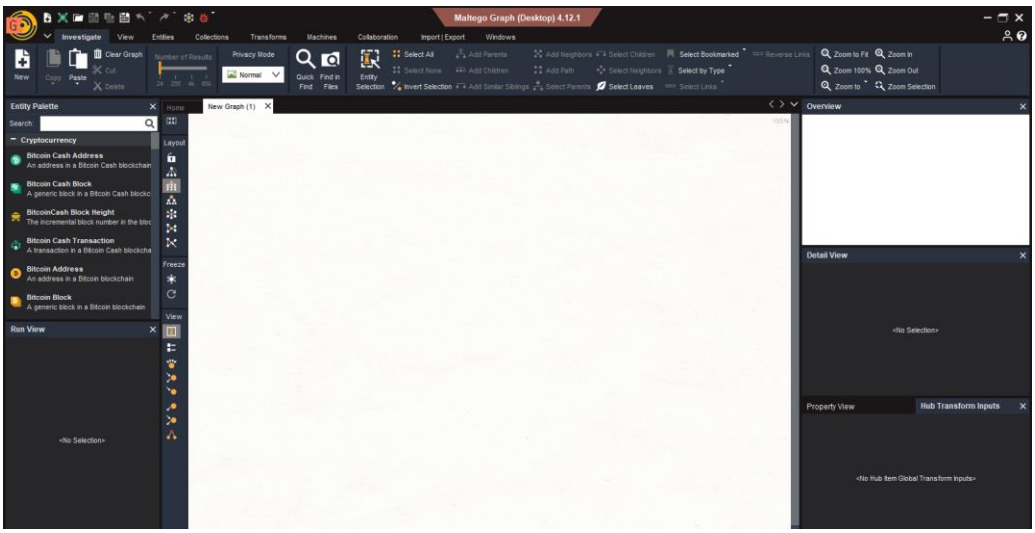
4.Investigate / Machines tab — run bundled, multi-step transform sequences automatically.



5. Entity Mapping — Starting an Investigation

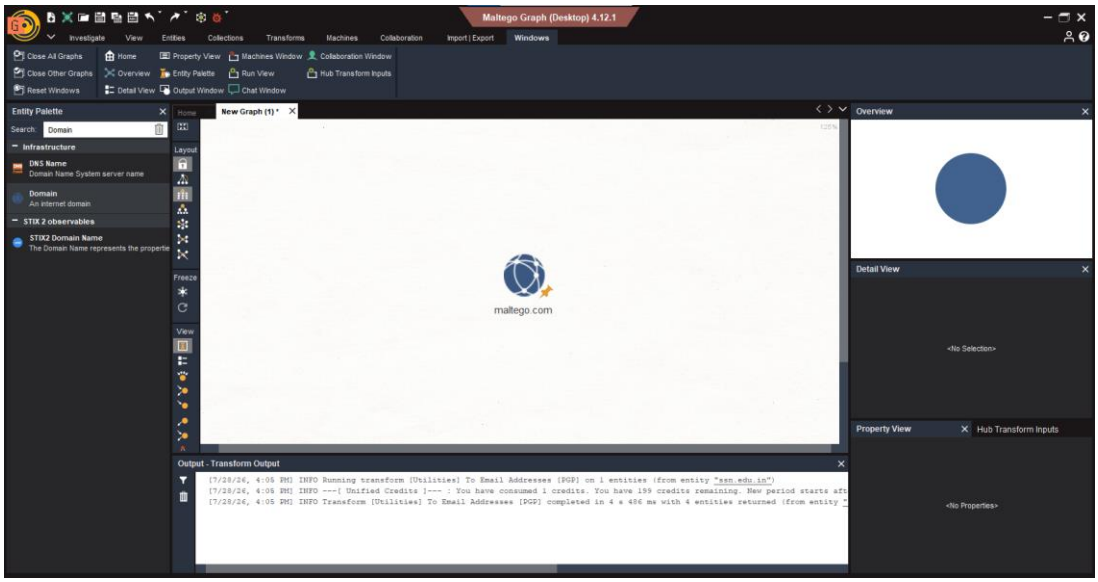
Step 1: Create a New Graph

Go to File → New Graph to open a blank investigation canvas.



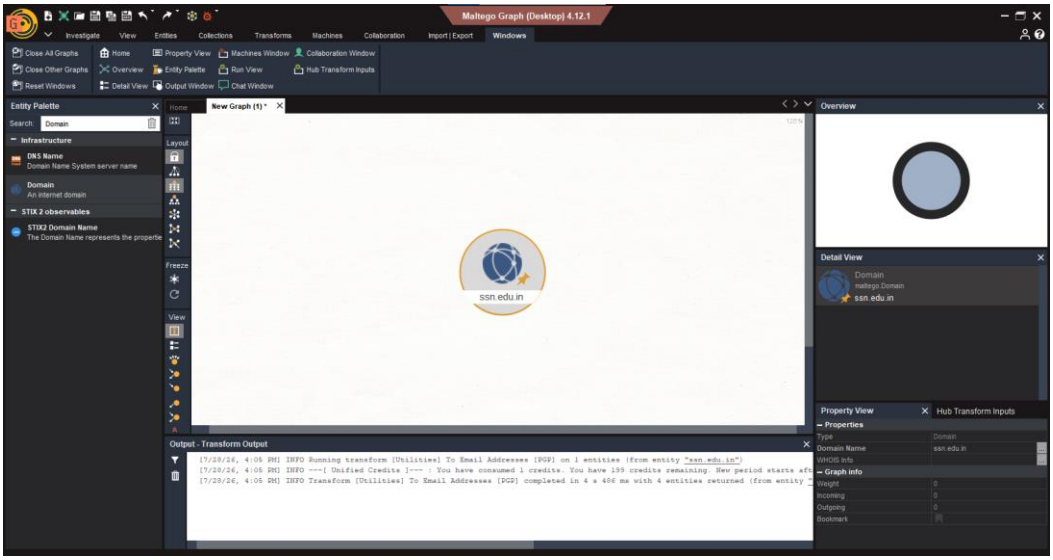
Step 2: Add a Domain Entity

Drag a “Domain” entity from the palette onto the canvas.



Step 3: Set the Target Domain

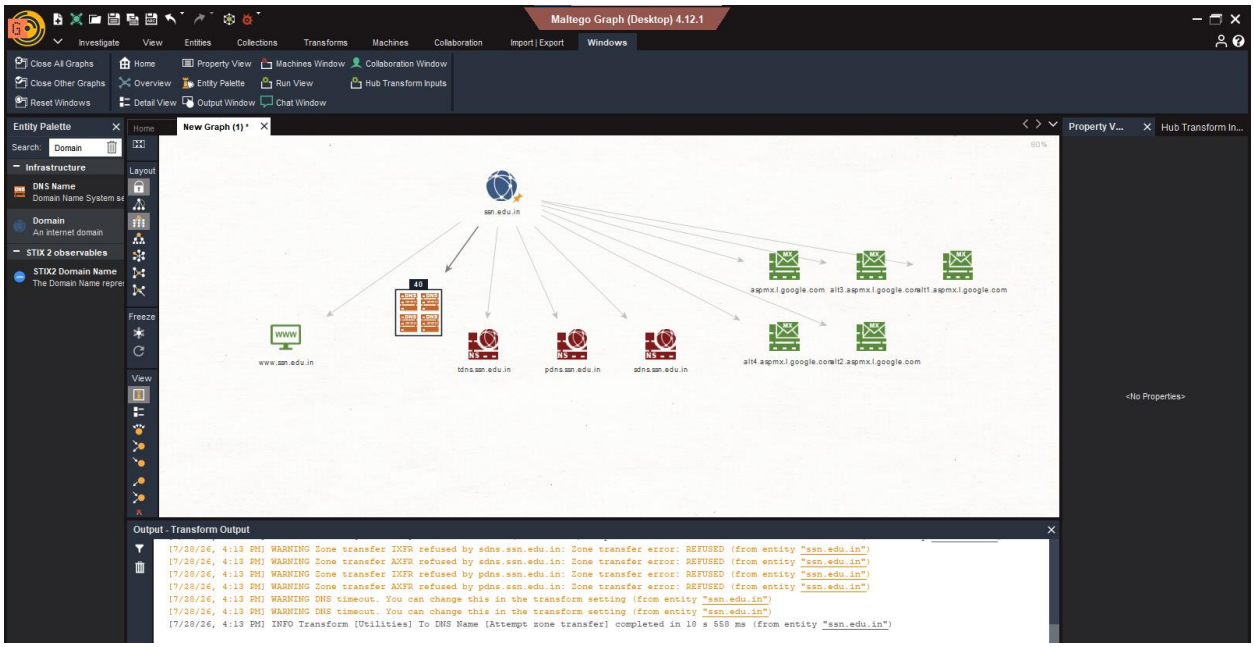
Double-click the entity and type your assigned training domain (e.g. a domain you own or an authorised test target).



6. Running Transforms — Expanding the Graph

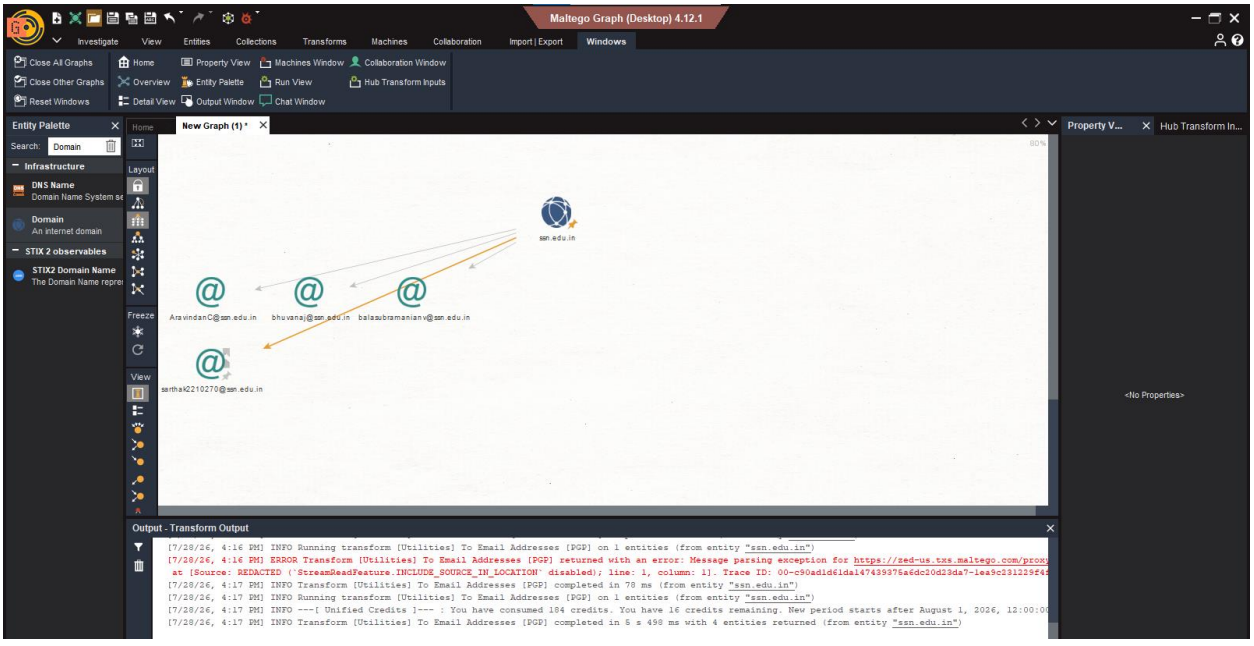
Step 1: Run a DNS Transform

Right-click the Domain entity → Run Transform → "To DNS Name – [DNS]" to discover linked subdomains.



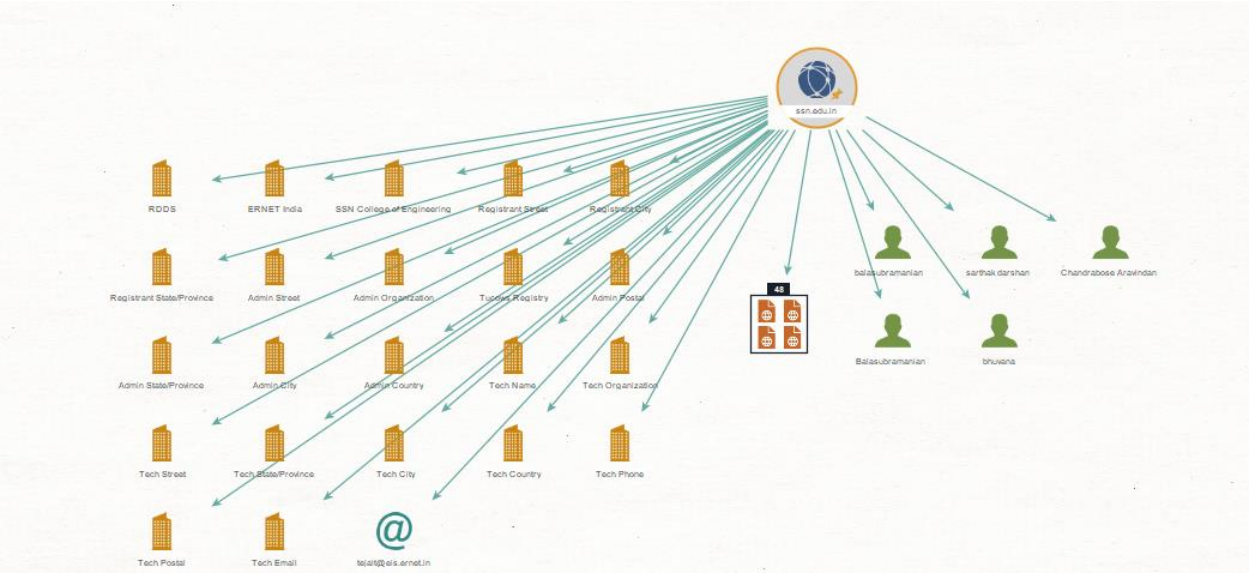
Step 2: Run an Email Transform

Right-click the Domain entity again → Run Transform → "To Email Address [SEC/PGP]" (or similar) to discover related email addresses.



Step 3: Resolve IP Addresses

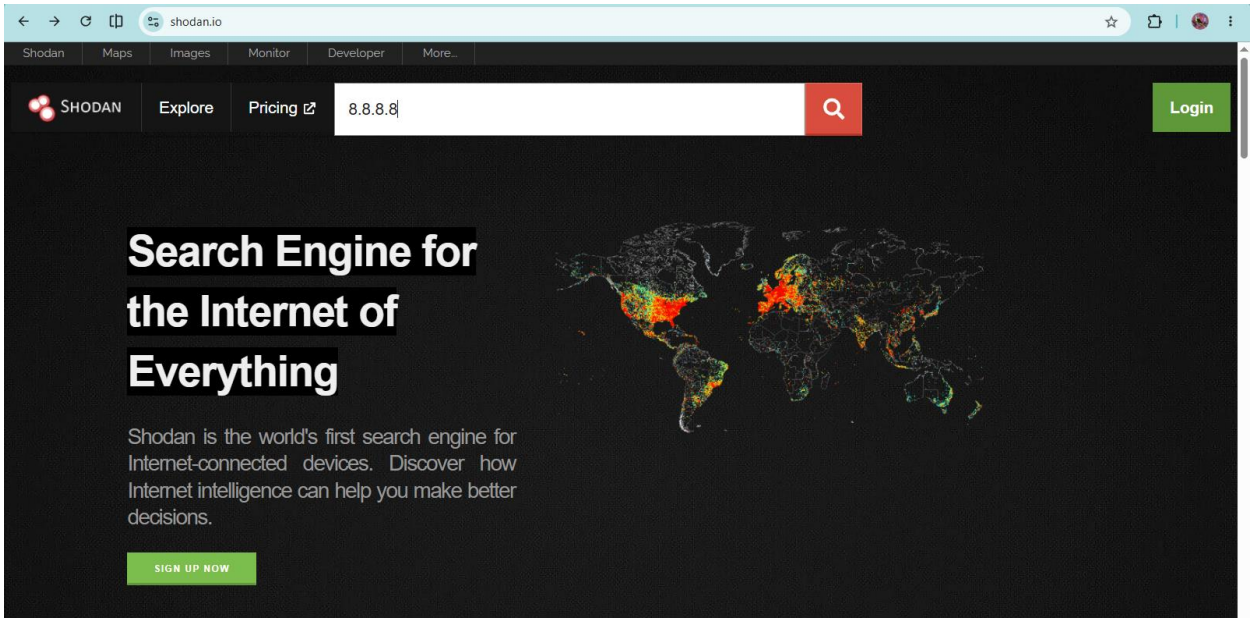
Right-click a discovered DNS Name entity → Run Transform → "Persons from Domain" to resolve it to an users.



7. Using Shodan to Enrich Findings

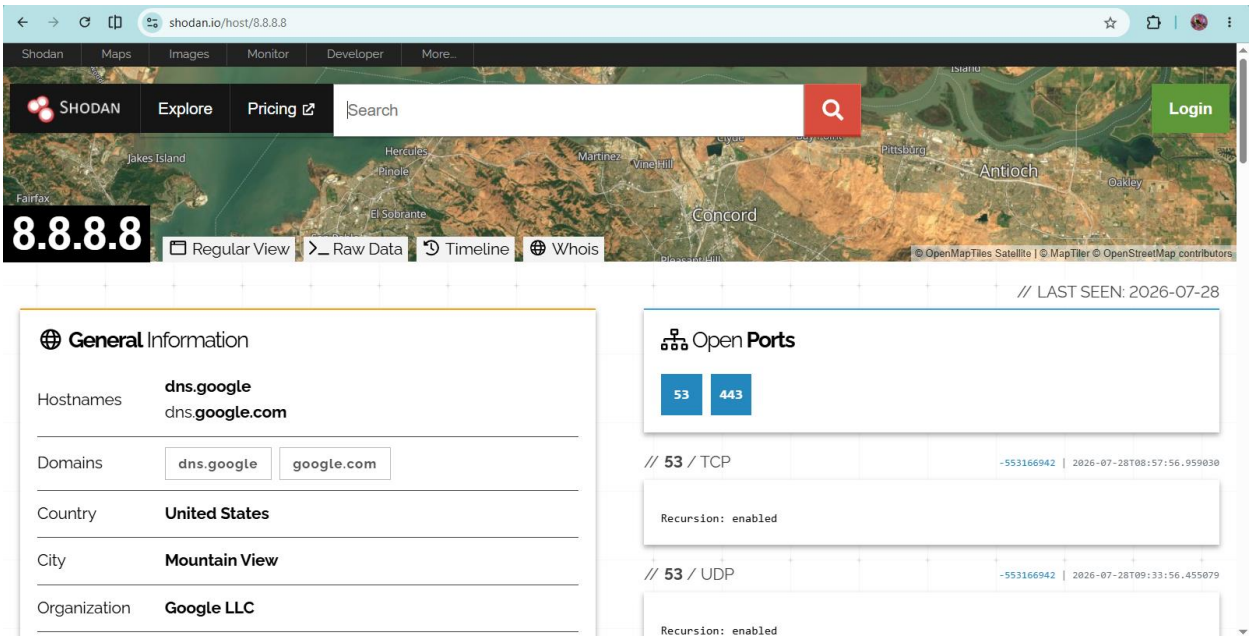
Step 1: Select an IP Address Entity(8.8.8.8)

Choose one IP Address entity discovered in your Maltego graph.



Step 2: Search the IP on Shodan

Visit shodan.io and search for the selected IP address to view any publicly observed open ports and service banners.



Step 3: Record the Findings

Note the open ports, service banners, and any location data shown.